ITA0328 - Mobile computing

Resource Management strategies in cellular Networks

P. Sathish.
192312182.

Goal

Efficient use of limited radio resources for higher capacity better coverage.

Channel Allocation

- Assign time / frequency channels to user or cells
- Improves spectral efficiency and reduces blocking.

Frequency reuse

- Allows same frequencies to be used in different cells.
- Needs proper planning to limit interference.

Power control

- Adjusts transmission power of each user / cell.
- Reduces co-channel and inter-cell interference.

Interference Management

- uses techniques like power control, beamforming, coordination, etc.
- Improves SINR and network

Handoff Management.

- Transfers a user to a new cell without service interruption.
- uses signal strength, quality and load.

works with

Supports by reuse

ensure stable links for handoff.

needs stable signal

Reduces interference for better

Contribution to Network utilization:

Channel Allocation	Frequency Reuse	Power control.	Interference Management	Handoff Management.
+ Efficient use of time / frequency resources prevents blocking Moderate impact.	+ Increases total capacity and coverage High impact.	+ + + directly reduces interference and enable higher reuse and capacity Highest impact.	+ + Maintain good SINR and service qual High impact	Keep users connected and balances load across cells. Moderate to high.

Conclusion

Power control contributes the most to efficient network utilization. and enables higher frequency reuse and overall capacity